The Second Corpus Study on German Nominal Compounding Constraints

Background

German nominal compounding is a notoriously unpredictable and irregular process. About 40% of German N+N compounds insert a so-called linking element that is not confidently predictable for most of constituent classes. For example, nouns Birne ‘pear’ and Kirsche ‘cherry’ build compounds Birne-n-saft ‘pear juice’ but Kirsch-0-saft ‘cherry juice’ despite having nearly identical phonological, morphological, and semantic properties. The only linguistic generalizations available to gauge the correct linker are (mostly) irregular constraints that are eagerly violated by numerous compounds.

Our first corpus study on German nominal compounding collected 64 constraints from the linguistic literature and verified 39 of them against ~280k German N+N compounds. The corpus study demonstrated that most of the constraints were irregular, indeed. The most crucial funding was that not all of the constraints were confirmed on the massive data. Specifically, eight constraints were only partially supported by the data, and three were not supported at all.

The main limitation of the first corpus study was that none of the constraints involving semantic or lexical conditions could be implemented. The present study fills in most of the gaps remaining after the first corpus study:

1. It implements 17 constraints involving semantics. Fourteen of the 17 constraints are pluralic constraints actively discussed in recent linguistics literature.
2. It applies more detailed statistic analyses to analyze relations between different ontological properties of the constraints (application manner, types of linguistic conditions etc.) and its coverage and regularity measures.
3. Additionally, it implements a couple of further constraints and verifies the constraints on further datasets.

Datasets

GeCoDB_v06

> 282767 compound lemmata. 3614 N1 lemmata. 9 linkers, 11 linker allomorphs. 4768 unique N1+linker combinations.

~280k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Database/compounds/decow16ax_comps_counts.txt'), with further processing steps in Shmalts BA Thesis, final preprocessing in Shmalts 2026.

The dataset is used to (re-)verify 40 non-semantic constraints.

GeCoDB_v07

> 275411 compound lemmata. 3593 N1 lemmata. 10 linkers, 12 linker allomorphs. 4741 unique N1+linker combinations.

~275k German N+N compounds from *GeCoDB_v06* whose constituents are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06*.

GeCoDB_v07_GN

> 48976 compound lemmata (48660 unique N1+N2 combinations). 3319 N1 lemmata. 10 linkers, 12 linker allomorphs. 3585 unique N1+linker combinations.

~49k German N+N compounds from *GeCoDB_v07* that are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06* and five further semantic non-pluralic constraints.

SP18_Pl_Constr

> 2294 compound lemmata. 43 N1 lemmata. 8 linkers, 9 linker allomorphs. 76 unique N1+(Sg/Pl meaning) combinations. 86 unique N1+linker combinations.

~2.3k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo, with annotated singular/plural meaning of N1. The dataset originates from Schäfer & Pankratz 2018 (file

'linkingelements/Data/Corpusstudy/Concordance.full.csv').

The dataset is used to verify 12 pluralic constraints.

Outline

1. Chapter 1. We first inspect the ontological properties of constraints as linguistic generalizations, as opposed to their empirical application in the language: application manner, type(s) of accessed linguistic properties, application item. The results corroborate and slightly elaborate the finding from the first corpus study.
2. Chapter 2. We then inspect the impact of data and metric choice on surface constraint applicability. We find that the constraint applicability values are statistically very similar both across the *GeCoDB* dataset versions and the item/type variants of the metrics. Based on these results, we focus on the results of *GeCoDB_v07_GN* and *SP18_Pl_Constr* and only concentrate on the item variant of the applicability statistics.
3. Chapter 3. We analyze the values of application measures of the constraints on the flattened applicability statistics, which includes their distribution and correlation. We establish relation between the values of coverage and regularity of the constraints, and their ontological properties from step 1.
4. Chapter 4. We finally introduce and apply the new (irregularity) scale to our results.

Results

1. Constraint Ontological Properties

The results corroborate and slightly elaborate the finding from the first corpus study.

- Morphology constitutes the core of German nominal compounding, followed by derivation and phonology. Semantics, while contributing to a considerable number of constrains, plays a complementary role and is almost negligible when isolated. Lexical and most of mixed constraints are underrepresented.
- N1 stays the cornerstone of constraint definitions. Al but two out of 65 constraints do not access the linguistic properties of N1.

2. Constraint Applicability Vs. Datasets & Metrics

- The values of both item and type coverage and regularity come from an identical distribution and are highly correlated across all pairwise comparisons of *GeCoDB* versions (Wilcoxon signed-rank test all p-values > 0.1). That speaks in favor of the validity of the original results. Additionally, it attests the sufficient data quality and ensures the robustness of the constraint applicability statistics. We choose to proceed with *GeCoDB_v07_GN* as the dataset with the most verified constraints.
- The item and type variants of both applicability metrics are statistically very similar (Wilcoxon signed-rank test both p-values > 0.1), which supports the equivalent finding from the first corpus study. We choose to proceed with the item variant as the more balanced version of the metrics.

3. Constraint Applicability

- Most of the item coverage values are very low at <0.05. The item regularity values are more even. They are clustered naturally by the thresholds of 0.7 and 0.9, which appear to cut through the first and the third largest gaps between two neighboring item regularity values. We propose a new (ir)regularity scale of $[0.5; 0.7)$, $[0.7; 0.9)$, $[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints (Mann-Whitney U-test p-value 0.001, permutation test p-value 0.002), while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho -0.177 and -0.337, respectively).
- We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff -0.21 at intercept 0.86, p-value 0.001), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

4. Constraint Support, Productivity & Regularity

- The new (ir)regularity scale did not affect the support level of previously verified constraints much: only four constraints from Shmalts 2026 downgraded to partially support at the new scale. The new *GeCoDB* versions produce an even better result on the old constraints due to cleaner data. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*. On top of that, the changed checkers

for the constraints triggered by morphological and phonological complexity of the N1 led to these two constraints being partially supported and unsupported, respectively.
- Ten constraints out of 48 turn out to be unproductive by the Tolerance Principle. Regular constraints by the ITP constitute half of the constraint set: 24 out of 48.
- We find that no constraints require a change of definition. Only regularity levels should be adjusted for them.

3. Constraint Applicability

Description

This chapter is the core part of the notebook. It analyses the applicability values of the constraints, which includes their distribution and correlation, and seeks to establish a relation between the values of coverage and regularity of the constraints, and their ontological properties considered in chapter 1.

1. First, we qualitatively analyze the distribution of the applicability values of the constraints. We observe a natural clustering of the item coverage and regularity values, and propose a new (ir)regularity scale based on it.
2. We compare the item coverage and regularity values of the P2L vs. L2P constraints. The L2P constraints appear to be statistically more regular. The difference in coverage is unexpectedly insignificant.
3. We test for a correlation between coverage and regularity. We find that there exists a small negative correlation between both item and type coverage and regularity.
4. We investigate the effect of the domain type of constraints on their item regularity. We find that semantics is the only domain that has a significant effect on item regularity. As expected, the effect is negative.

NB! In this chapted, we filter out all unconfirmed constraints (item regularity of $<0.5$$<0.5$) and constraints with $<3$$<3$ covered items.

Results

- Most of the item coverage values are very low at $<0.05$$<0.05$. The item regularity values are more even. They are clustered naturally by the thresholds of $0.7$$0.7$ and $0.9$$0.9$, which appear to cut through the first and the third largest gaps between two neighboring item regularity values. We propose a new (ir)regularity scale of $[0.5; 0.7)$$[0.5; 0.7)$, $[0.7; 0.9)$$[0.7; 0.9)$, $[0.9; 1.0]$$[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints (Mann-Whitney U-test p-value $0.001$$0.001$, permutation test p-value $0.002$$0.002$), while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho $-0.177$$-0.177$ and $-0.337$$-0.337$, respectively).
- We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.21$$-0.21$ at intercept $0.86$$0.86$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

First, we only keep constraints that have at least $3$$3$ covered items and an item regularity of $\geq 0.5$$\geq 0.5$. That results in $48$$48$ constraints. Again, the number of the P2L constraints almost doubles the number of the L2P constraints.

Start coding or generate with AI.

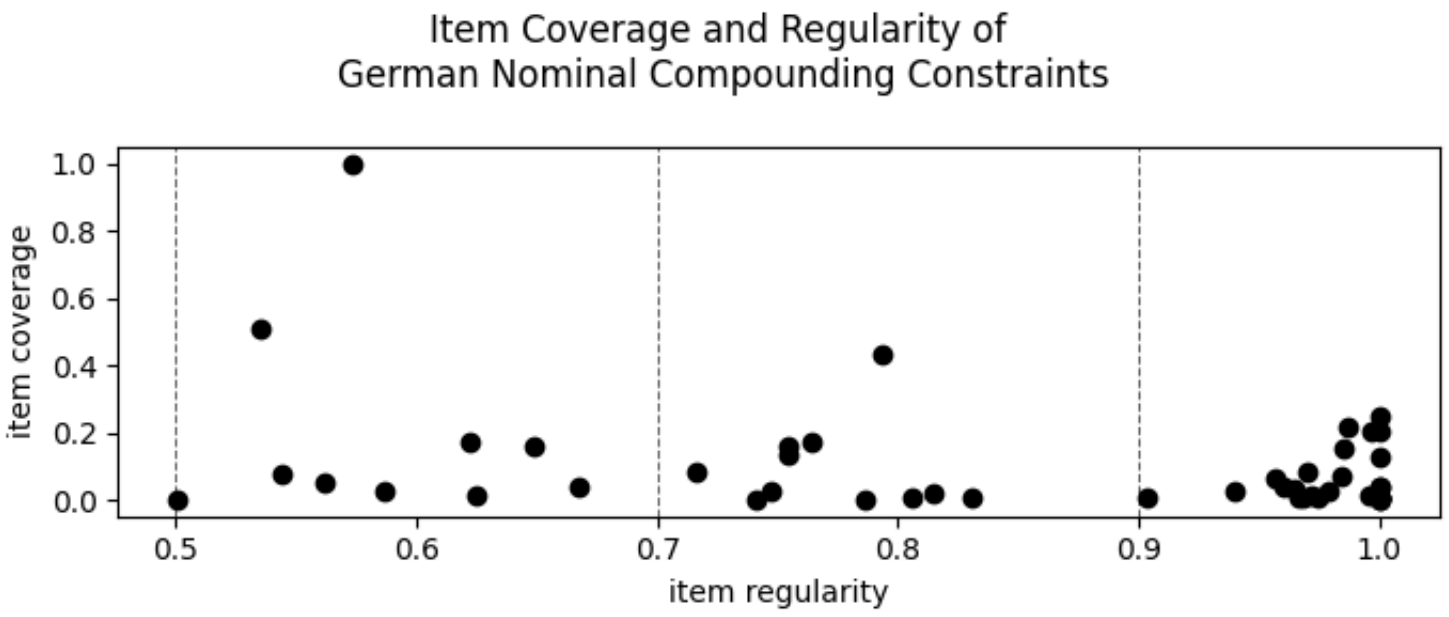
	N
total	48
p2l	31
l2p	16

1. As already reported in Shmalts 2026, the item coverage values are mostly very low, with more than half of the constraints lying in the item coverage range of $[\sim 0.0; 0.05)$$[\sim 0.0; 0.05)$ (28 of 48, or $53.3\%$$53.3\%$).

What concerns the item regularity values, the present study produces novel results. Specifically, it seems that the thresholds of $0.7$$0.7$ and $0.9$$0.9$ are the natural barriers between "low", "moderate", and "high" regularity. Indeed, the respective clusters are seen clearly on the scatter plot. The legitimacy of this values is further supported by the fact that they cut through the first and the second largest gaps between neighboring item regularity values in the dataset.

The item regularity values clustering calls the original regularity scale from Shmalts 2026 into question. We propose the newly discovered thresholds of $0.7$$0.7$ and $0.9$$0.9$ as the new (ir)regularity scale. More on completion of the new scale with a "regular" level follows in chapter 4.

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	N
cvg: [0.0-0.05)	28
cvg: [0.05-0.2)	13
cvg: [0.2-1.01)	7
reg: [0.5-0.7)	10
reg: [0.7-0.9)	11
reg: [0.9-1.01)	27

The two largest gaps betwee neighboring item regularity values are $0.831$$0.831$-$0.903$$0.903$ and $0.667$$0.667$-$0.716$$0.716$. They are cut precisely by the thresholds of $0.9$$0.9$ and $0.7$$0.7$, respectively.

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	left	right	diff
0	0.831	0.903	0.072
1	0.667	0.716	0.049
2	0.903	0.940	0.037
3	0.587	0.622	0.035
4	0.501	0.535	0.034
5	0.716	0.741	0.025
6	0.625	0.649	0.024
7	0.764	0.786	0.022
8	0.544	0.562	0.018
9	0.649	0.667	0.018

2. The L2P constraints appear to be statistically more regular than the P2L constraints. This conclusion corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirm its robustness after adding semantic constraints. On the contrary, there was no observed statistical difference between the item coverage values of the P2L vs. the L2P constraints, as unexpected as it was. This finding contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.

Almost all L2P applicability values have a very low coverage and a high regularity, while the applicability values of the P2L constraints are distributed more or less evenly. Thus, 12 out of 16, or $75\%$$75\%$ of the L2P constraints have an item coverage $<0.05$$<0.05$. For the P2L constraints, this number is only $51.6\%$$51.6\%$. The item regularity values are concentrated much closer to $1.0$$1.0$ among the L2P but not the P2L constraints. There are none L2P constraints with item regularity $<0.7$$<0.7$. On the contrary, there are nine, or $29\%$$29\%$, such P2L constraints. Almost all L2P constraints are highly regular ($\geq 0.9$$\geq 0.9$): 13 out of 16, or $81.3\%$$81.3\%$. The P2L constraints do not even come close with mere $45.2\%$$45.2\%$ of highly regular constraints.

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	all	P2L	L2P
cvg: [0.0-0.05)	28	16	12
cvg: [0.05-0.2)	13	11	2
cvg: [0.2-1.01)	7	4	2
reg: [0.5-0.7)	10	9	0
reg: [0.7-0.9)	11	8	3
reg: [0.9-1.01)	27	14	13

We now run a Mann-Whitney U-test over the applicability values of the P2L vs. the L2P constraints. We choose the Mann-Whitney U-test because we are dealing with two independent samples whose distributions are abnormal and skewed. Since we suppose that the L2P constraints have lower coverage and higher regularity, we choose the two one-tailed alternative hypotheses: greater and less.

We can clearly see that the L2P constraints are undeniably more regular than the P2L constraints at sig = 0.001 (p-value $0.001$$0.001$; one-tailed alternative: L2P item regularity values are greater). They also seem to have a lower coverage, with p-value of $0.021$$0.021$ (one-tailed alternative: P2L item coverage values are greater).

Start coding or generate with AI.

		cvg	reg
alternative			
P2L>L2P	U	338.500	115.000
	p	0.021	0.999
P2L<L2P	U	338.500	115.000
	p	0.980	0.001

Since there are a few outliers in the P2L item coverage values, and *SP18_Pl_Constr* is a small dataset that can be not representative as of coverage, the results of the Mann-Whitney U-test can biased. We therefore additionally run a permutation test below. It compares the values themselves and not their ranks and is therefore much more sensitive to the outliers.

The permutation test confirmed that the item regularity values are higher among the L2P constraints (observed shift of the P2L vs. the L2P item regularity means $-0.135$$-0.135$, p-value $0.002$$0.002$). It also disperses our doubts about item coverage: the difference is not statistically different, after all, with p-value of $0.091$$0.091$.

Start coding or generate with AI.

		cvg	reg
alternative			
P2L>L2P	shift	0.045	-0.135
	p	0.091	0.998
P2L<L2P	shift	0.045	-0.135
	p	0.916	0.002

To demonstrate the bias of the outliers, we run a dedicated Mann-Whitney U-test over *GeCoDB_v07_GN* only with an item coverage threshold og $0.5$$0.5$, which is still much. While the robust regularity difference stays significant at p-value $0.003$$0.003$, the significance of the coverage difference vanishes (p-value $0.072$$0.072$). We conclude that there is no difference in item coverage between the P2L and the L2P constraints, after all.

Start coding or generate with AI.

		cvg	reg
alternative			
P2L>L2P	U	235.500	89.000
	p	0.072	0.997
P2L<L2P	U	235.500	89.000
	p	0.932	0.003

3. Contradictory to the claim in Shmalts 2026, there exists a (very) small negative correlation between both item and type coverage and regularity (Spearman rho $-0.177$$-0.177$ and $-0.337$$-0.337$, respectively). Statistically, constraints with lower coverage have slightly higher regularity. This is in line with our assumption that the more specific constraints are stronger than the general ones. The large difference in the correlation coefficient between the item and type variants is a result of different productivity of items what regards building compounds.

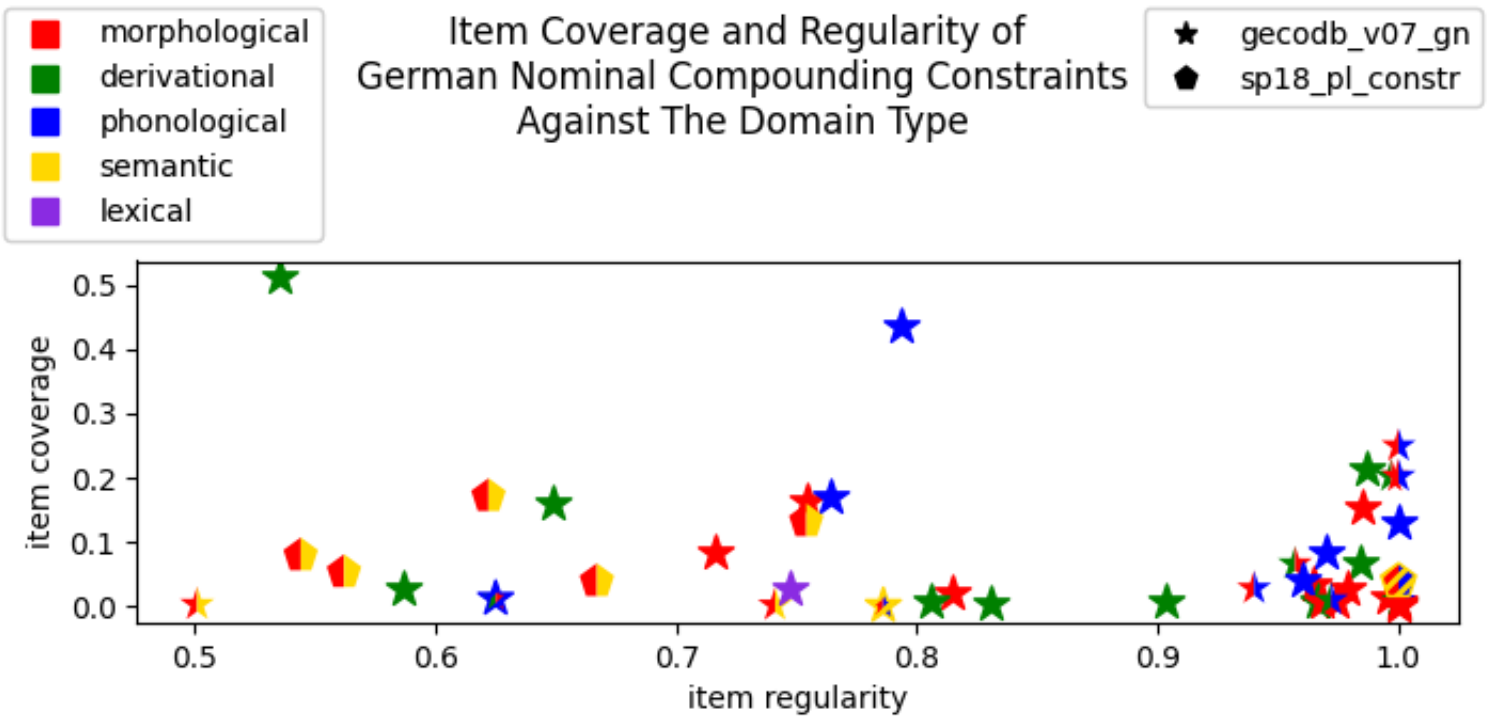
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	item	type
r	-0.177	-0.337

4. We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding. Note that all non-confirmed constraints (item regularity $<0.5$$<0.5$) are semantic as well which makes the real picture even worse for semantics.

Let us consider the scatter plot of constraint item regularity against the domain type. We can immediately see that morphological, derivational, and phonological constraints are scattered all over the lot but are still concentrated towards the right edge of the axis. Contrary to that, semantic (including mixed) constraints only populate the range of $[0.5; 0.8]$$[0.5; 0.8]$; the only constraint with a high regularity is morphological+phonological+semantic. The plot suggests that the constraints that include a semantic condition are less regular.

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To confirm the intuition, we run a linear regression over the item regularity of the constraints vs. their domain type, based on the one-hot vectors we produced above. To concentrate on individual contribution of each domain, we do not include their interaction terms.

The linear regression model confirms our intuition. The model overall model is significant and explains more variation of item regularity than chance (p-value $0.012$$0.012$). In the model, semantics is the only domain that contributes to item regularity significantly (coeff $-0.21$$-0.21$ at intercept $0.86$$0.86$, p-value $0.001$$0.001$). That is, the item regularity of an arbitrary constraint decreases by $0.21$$0.21$ when a semantic condition included, assuming that the other domains of this constraint remain intact. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

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```
                            OLS Regression Results
==============================================================================
Dep. Variable:                reg_item   R-squared:                       0.293
Model:                            OLS   Adj. R-squared:                  0.207
Method:                 Least Squares   F-statistic:                     3.406
Date:                Fri, 04 Sep 2026   Prob (F-statistic):             0.0115
Time:                        18:46:51   Log-Likelihood:                 27.678
No. Observations:                  47   AIC:                            -43.36
Df Residuals:                      41   BIC:                            -32.25
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept                  0.8606      0.065     13.168      0.000       0.729       0.993
morphological[T.True]      0.0546      0.062      0.881      0.384      -0.071       0.180
derivational[T.True]      -0.0256      0.062     -0.415      0.680      -0.150       0.099
phonological[T.True]       0.0684      0.050      1.361      0.181      -0.033       0.170
semantic[T.True]          -0.2112      0.058     -3.660      0.001      -0.328      -0.095
lexical[T.True]           -0.1136      0.158     -0.720      0.476      -0.433       0.205
==============================================================================
Omnibus:                        2.818   Durbin-Watson:                   1.474
Prob(Omnibus):                  0.244   Jarque-Bera (JB):                2.379
Skew:                          -0.550   Prob(JB):                        0.304
Kurtosis:                       2.928   Cond. No.                         10.2
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

If we include the interaction terms, the picture stays the same. Now morphology, derivation, phonology, and semantics (but not the lexical domain) significantly contribute to item regularity (p-value in $[0.0; 0.001]$$[0.0; 0.001]$). From these four isolated domains, only semantics has a negative coefficient ($-0.128$$-0.128$) which the former three increase item regularity by $0.274$$0.274$, $0.174$$0.174$, $0.196$$0.196$, respectively (intercept $0.672$$0.672$). From all the interaction terms, the interaction of morphology and semantics is significant (coefficient $-0.128$$-0.128$, p-value $0.0$$0.0$). Otherwise, the interaction morphology+derivation+phonology is significant, too (coefficient $-0.352$$-0.352$, p-value $0.034$$0.034$) but we discard it since there is only one such constraint, which cannot assure statistical power.

The domain type of constraints has no effect on their item coverage (p-value of the overall model $0.741$$0.741$, none of the individual domains are significant).

Start coding or generate with AI.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                cvg_item   R-squared:                       0.062
Model:                            OLS   Adj. R-squared:                 -0.052
Method:                 Least Squares   F-statistic:                    0.5457
Date:                Fri, 04 Sep 2026   Prob (F-statistic):              0.741
Time:                        18:59:05   Log-Likelihood:                 39.408
No. Observations:                  47   AIC:                            -66.82
Df Residuals:                      41   BIC:                            -55.71
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept                  0.1307      0.051      2.567      0.014       0.028       0.234
morphological[T.True]     -0.0549      0.048     -1.137      0.262      -0.152       0.043
derivational[T.True]      -0.0398      0.048     -0.830      0.412      -0.137       0.057
phonological[T.True]       0.0086      0.039      0.220      0.827      -0.070       0.088
semantic[T.True]          -0.0217      0.045     -0.484      0.631      -0.113       0.069
lexical[T.True]           -0.1047      0.123     -0.851      0.400      -0.353       0.144
==============================================================================
Omnibus:                       32.088   Durbin-Watson:                   0.994
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               68.875
Skew:                           1.977   Prob(JB):                     1.11e-15
Kurtosis:                       7.420   Cond. No.                         10.2
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```